Figure 3. Enlargement slows both turning routes and selectively suppresses edge susceptibility

A. Wave-pinning activity profiles for a cue straddling the incumbent edge (i) or separated from it (ii), C=30. Rows show the initial state, half-turn, 90% turn and cue-free relaxation. Grey marks cue support, diamonds mark saddles and the black tick marks activity heading.

B. Enlargement slows half-turning through both incumbent sliding and replacement by a new region. Half-turn time is plotted against cue-edge gap for C=30, 37.8, 47.6 (Spring 2.5, 3.15, 3.97), light to dark. Vertical lines mark the first saddle at half-turn; grey marks undefined heading at one or more sizes. Open upward triangles with arrows denote time lower bounds.

C. Signed terminal displacement per dose for a Gaussian pulse centred at each arc position, at f=0.001: SD 1.5, duration 50 (Spring 0.15, 5). Light and dark are C20/C60 (Spring 1/3). Lines are the native linearized response to this pulse shape; circles are direct pulse tests and open squares mark the edge test. The edge pulse confirms the map locally; additional tests cover the domain centre, halfway to the edge, edge ±1 and ±3 pulse SD, quarter-basal arc and antipode. Maximum absolute gain error is 1.69% of map amplitude. Largest/smallest terminal edge-gain folds are Wave-pinning 0.701, Holmes 0.914, Spring 0.989, Goryachev 0.088. Enlargement suppresses susceptibility strongly in Wave-pinning and Goryachev, while Holmes and Spring change little. The spatial response supplies the drift prediction in G.

D. Mean-state activity on the same centred physical arc coordinate as C and E. Dotted lines locate half-contrast edges. Spring uses material positions multiplied by mean circumference.

E. Nearly all predicted drift (99.6–100%) is generated within two input correlation lengths of the two edges, suggesting lateral-edge fluctuations as the cellular readout. Curves show signed per-position contributions divided by total predicted diffusion; negative values remain. Rounded shares per edge/plateau interior/basal arc (%) are: Wave-pinning C20: 50.0/<0.1/<0.1; Wave-pinning C60: 49.9/0.1/<0.1; Holmes C20: 50.0/<0.1/<0.1; Holmes C60: 50.0/<0.1/<0.1; Spring C1: 50.0/0/0; Spring C3: 49.8/0.2/0.2; Goryachev C20: 50.0/0/<0.1; Goryachev C60: 50.0/0/<0.1. Interior and basal zones exclude edge zones; exact signed fractions are supplied with the source data.

F. Wave-pinning C20 and C60: the pulsed edge (solid) and opposite edge (dashed) move together and retain a permanent shift after withdrawal. Grey marks pulse-on; displacement is divided by pulse dose. Other time courses and integrated responses are supplementary.

G. A weak input anywhere shifts the domain by g(x) times dose; noise is many such inputs; drift follows from g and the input covariance; nothing is fitted. Shared-translation predicted versus measured diffusion is divided by each quantity's own smallest-size value at the same input (domain drift is also shown in Figure 4G). Grey is equality. Colours encode size; shapes encode f=0.0001, 0.0003, 0.001, 0.003, 0.005, 0.01 (circle, square, upward triangle, downward triangle, diamond, filled plus). Horizontal intervals are measured 95% bootstrap intervals. Wave-pinning's terminal gain fold gives $0.70^2 \approx 0.49$; the full spatial covariance predicts 0.490, versus 0.473 measured at f=0.001. The square relation is approximate because spatial shape and covariance also change. Goryachev fails quantitatively and retains its own scale. Absolute diffusion and the independent-edge candidate are supplementary.

Figure 3. Enlargement slows both turning routes and selectively suppresses edge susceptibility · continued

H. The drift prediction implies a local size exponent, compared with measured heading-persistence α from Figure 4E. Predictions use the exact mean input of every observation; no f=0.001 extrapolation is used. The drift subset requires f≤0.01 and multiple-domain fraction≤0.05 throughout the fixed size neighbourhood. Native stationary-state and pulse-linearity gates must also pass. Grey is equality; vertical intervals are observed 95% bootstrap intervals; colours and shapes match G. 53 qualified pairs include all 15 Wave-pinning observations at f=0.003, 0.005 and 0.01. Its predicted α spans 2.48–2.88, above the size-independent-drift expectation of 2. Within Wave-pinning, this comparison tests the excess α above 2 more than the within-model trend. Goryachev has no qualified pair. Every excluded condition and reason remains in the source data.

$$G(x_0) = \sum_j g_j h(x_j - x_0) \tag{3.1}$$

The main map is the standard Gaussian-pulse response per dose, not raw cell-input g.

$$c_j = g_j \sum_k g_k \int_0^\infty \overline{K_u(j,k;t,t+v)}^{\,t}\, \mathrm{d}v, \qquad D_s = \sum_j c_j \tag{3.2}$$

Signed per-position contributions retain the exact native input covariance.

$$\alpha_{\mathrm{pred}} = 2 - \frac{\mathrm{d}\ln D_s}{\mathrm{d}\ln \bar{C}} \tag{3.3}$$

Same fixed adjacent-size neighborhoods as Figure 4E; exact matched-input predictions and explicit drift-regime eligibility.